25901735 **A**

Reg.No : .....

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION APRIL 2026

FOURTH SEMESTER

Multi-Disciplinary Courses (MDC) - MG2MDCMAT100 - APPLICABLE MATHEMATICS

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 15 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple type Questions

Answer any **25** questions

Each question carries **2** marks

1. Find BA if $A = \begin{bmatrix} 1 & 2 \\ 2 & 3 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & -2 & 1 \\ 0 & 1 & -2 \\ 3 & -4 & 5 \end{bmatrix}$. [A] / [CO1]

a). $\begin{bmatrix} 0 & 4 \\ -4 & -5 \\ 10 & 14 \end{bmatrix}$ b). $\begin{bmatrix} 2 & 4 \\ -4 & -5 \\ 10 & 14 \end{bmatrix}$

c). $\begin{bmatrix} 2 & 4 \\ 4 & 5 \\ 10 & 14 \end{bmatrix}$ d). $\begin{bmatrix} 2 & 4 \\ 4 & 5 \\ 10 & 10 \end{bmatrix}$

2. Find $A+B$ if $A=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$ and $B=\begin{bmatrix}0 & 0\\ 0 & 1\end{bmatrix}$. [A] / [CO1]

a). $\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$ b). $\begin{bmatrix}2 & 0\\ 0 & 2\end{bmatrix}$

c). $\begin{bmatrix}1 & 0\\ 0 & 3\end{bmatrix}$ d). $\begin{bmatrix}1 & 0\\ 0 & 2\end{bmatrix}$

3. Which of the following is a diagonal matrix? [U] / [CO1]

a). $\begin{bmatrix}0 & 2 & 1\\ 2 & 0 & 1\\ 2 & 1 & 0\end{bmatrix}$ b). $\begin{bmatrix}5 & 1 & 0\\ 0 & 5 & 0\\ 0 & 0 & 5\end{bmatrix}$

c). $\begin{bmatrix}4 & 0 & 0\\ 0 & 5 & 0\\ 0 & 0 & 9\end{bmatrix}$ d). $\begin{bmatrix}2 & 2 & 2\\ 3 & 3 & 3\\ 4 & 4 & 4\end{bmatrix}$

4. Which of the following is true about matrix subtraction? [U] / [CO1]

a). It is always commutative.

b). It can only be performed if the matrices have the same order.

c). The result is always a diagonal matrix.

d). Subtracting any matrix from itself gives the identity matrix.

5. Find a_{32} from the matrix $A=\begin{bmatrix}2 & 1 & 0\\ 0 & -1 & 3\\ 4 & 3 & 2\end{bmatrix}$ [U] / [CO1]

a). 2

b). 3

c). 4

d). 0

6. If A is $\begin{bmatrix}-2i & 3i\\ 2 & 5\end{bmatrix}$, the conjugate transpose of A is [A] / [CO1]

a). $\begin{bmatrix}2i & -3i\\ 2 & 5\end{bmatrix}$ b). $\begin{bmatrix}2i & 2\\ -3i & 5\end{bmatrix}$

c). $\begin{bmatrix}2i & 2\\ 5 & -3i\end{bmatrix}$ d). $\begin{bmatrix}-2i & 3i\\ 5 & 2\end{bmatrix}$

7. If A and B are symmetric matrices of the same order, then $A+B$ is_____ [U] / [CO1]

a). Always symmetric

b). Always skew-symmetric

c). Symmetric or skew-symmetric d). Always a diagonal matrix

8. Find $2A$ if $A = \begin{bmatrix} 0 & 1 & 2 \\ 2 & 3 & 4 \\ 4 & 5 & 6 \\ 5 & 6 & 7 \end{bmatrix}$ [A] / [CO1]

a). $\begin{bmatrix} 0 & 2 & 4 \\ 4 & 6 & 8 \\ 8 & 10 & 12 \\ 10 & 12 & 14 \end{bmatrix}$ b). $\begin{bmatrix} 0 & 2 & 5 \\ 3 & 6 & 8 \\ 8 & 3 & 12 \\ 9 & 12 & 14 \end{bmatrix}$

c). $\begin{bmatrix} 2 & 0 & 5 \\ 3 & 18 & 8 \\ 8 & 3 & 12 \\ 9 & 12 & 14 \end{bmatrix}$ d). $\begin{bmatrix} 2 & 4 & 0 \\ 4 & 6 & 8 \\ 8 & 3 & 12 \\ 9 & 12 & 14 \end{bmatrix}$

9. he factor the quadratic polynomial x^2-6x+9 is …… [K] / [CO2]

a). $x+3$ b). $x-1$

c). $x-9$ d). $x-3$

10. Which of the following is a cubic polynomial? [A] / [CO2]

a). $x^3+x^4+x^2+3$ b). x^3-2x+6

c). $1-3x+x^2$ d). $x+3$

11. What is the degree of the polynomial $(x+3)(4x+1)$ [A] / [CO2]

a). 2 b). 1

c). 4 d). 3

12. Which are the factors of $x^2-8x+15$? [K] / [CO2]

a). $(x+3),(x+5)$ b). $(x-3),(x+5)$

c). $(x+3),(x-5)$ d). $(x-3),(x-5)$

13. The roots of the quadratic equation $x^2-9x-52=0$ are [U] / [CO2]

a). 4,-13 b). -4,13

c). -4,-13 d). -4,11

14. The roots of the quadratic equation $x^2 + 20x + 99 = 0$ are [U] / [CO2]

a). $11, 9$ b). $-11, -9$

c). $1, 99$ d). $1, 20$

15. If r and n be two positive integers such that $r < n$ then $\frac{n!}{r!} =$ [A] / [CO3]

a). $n(n-1)....(n-r+1)$ b). $n(n-r+1)$

c). $n(n-1)....(r+1)$ d). $n(r+1)$

16. How many quadrilaterals can be formed by joining the vertices of an octagon? [A] / [CO3]

a). 32 b). 70

c). 12 d). 8

17. Evaluate $10 \times 3!$ [U] / [CO3]

a). 50 b). 60

c). 70 d). 80

18. The value of $\frac{6!}{5! \times 3!}$ is [U] / [CO3]

a). 0 b). 1

c). 2 d). 3

19. Total number of handshakes among the group of 42 men is [U] / [CO3]

a). 42 b). 84

c). ${}^{42}C_2$ d). ${}^{42}P_2$

20. In how many ways can 12 different books be divided equally among 3 sets or groups? [A] / [CO3]

a). ${}^{12}C_3$ b). $\frac{{}^{12}C_4 \times {}^{8}C_4 \times {}^{4}C_4}{3!}$

c). ${}^{12}C_4$ d). ${}^{12}C_4 \times {}^{8}C_4 \times {}^{4}C_4$

21. Find the number of words formed with the letters of the word 'DELHI' has the letter I always in the middle ? [A] / [CO3]

a). 14 b). 18

c). 22 d). 24

22. How many different numbers of four digits (with repetition of digits) can be formed from the digits 3, 1, 7, 9, 5? [A] / [CO3]

a). $5!^4$ b). 5^5

c). 5^4 d). None of the above

23. Find $\frac{dy}{dx}$ if $y = x\cos x$. [A] / [CO4]

a). $x\sin x + x\cos x$ b). $\sin x + \cos x$

c). $-x\sin x + \cos x$ d). None of these

24. Find the derivative of $y = (\tan x)^2$? [A] / [CO4]

a). $(\sec x)^2$ b). $2x\tan x$

c). $2\sec 2x$ d). $2\tan x\sec^2 x$

25. Find the derivative of $5(x^2 + 3x - 1)^4$ at $x = 0$ [A] / [CO4]

a). 0 b). -15

c). 15 d). 5

26. $Find\ the\ derivative\ of\ x\cos x$ [U] / [CO4]

a). $x\sin x - \cos x$ b). $\cos x$

c). $x\sin x$ d). $\cos x - x\sin x$

27. What is the value of $1 + \cos 0$? [U] / [CO4]

a). 1 b). 2

c). 0 d). 3

28. If $f(x) = \sqrt{x} + \frac{1}{x}$.then $f'(x)$ is [A] / [CO4]

a). $\frac{1}{2\sqrt{x}} - \frac{1}{x^2}$

b). $\frac{1}{\sqrt{2x}} - \frac{1}{x^2}$

c). $\frac{1}{\sqrt{2x}} - \frac{1}{x}$

d). $\frac{1}{2\sqrt{x}} + \frac{1}{x}$

29. $What\ is\ the\ derivative\ of\ f(x) = \frac{1}{(x+4)}$ [U] / [CO4]

a). $\frac{1}{(x+4)^2}$

b). $\frac{-1}{(x+4)^2}$

c). $(x+4)^2$

d). $\frac{4}{(x+4)^2}$

30. $\frac{d}{dx}(\sqrt{6x}) =$ [A] / [CO4]

a). $\sqrt{6}$

b). $\sqrt{\frac{3}{2x}}$

c). $\sqrt{\frac{2}{3x}}$

d). $\frac{\sqrt{6}}{2x}$

[2x25 = 50]

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b). $\begin{bmatrix} 5 & 1 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix}$

c). $\begin{bmatrix} 4 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 9 \end{bmatrix}$

d). $\begin{bmatrix} 2 & 2 & 2 \\ 3 & 3 & 3 \\ 4 & 4 & 4 \end{bmatrix}$

[2x25 = 50]